Enhanced Oil Recovery Project

Project Summary

The Enhanced Oil Recovery (EOR) Project aims to implement advanced enhanced oil recovery techniques in selected oilfields in Brazil to increase production from existing reserves. This project will target aging oilfields such as the Marlim and Lula fields to maximize oil extraction efficiency. The project is expected to significantly increase oil production, enhance energy security, and contribute to economic growth by creating jobs and generating revenue. Given the high risk level associated with the project, stringent safety protocols and comprehensive risk mitigation strategies will be implemented. The project is expected to yield a 22% Return on Investment (ROI), with financial benefits phased over time and shared between BP and Petrobras.

Project Scope of Work

Phase 1: Planning and Preparation

The initial phase involves site selection, detailed feasibility studies, procurement of EOR equipment, and obtaining regulatory approvals and permits. This phase is crucial for ensuring the project's compliance with environmental and safety regulations and for preparing the necessary infrastructure for EOR operations.

- Site Selection and Feasibility Studies (Sep 2023 - Dec 2023)
- Procurement of EOR Equipment (Dec 2023 - Feb 2024)
- Regulatory Approvals and Permits (Sep 2023 - Mar 2024)

Phase 2: Implementation and Execution

The implementation and execution phase includes the deployment of EOR techniques, monitoring of recovery processes, and optimization of production. This phase will utilize advanced EOR technology to maximize oil extraction efficiency.

- Deployment of EOR Techniques (Mar 2024 - Jun 2024)
- Monitoring and Optimization of Recovery Processes (Jul 2024 - Dec 2024)
- Continuous Improvement and Adjustments (Jan 2025 - Sep 2025)

Phase 3: Evaluation and Reporting

The final phase involves data analysis, interpretation of results, and the preparation of final reports and recommendations. This phase ensures that the project's outcomes are thoroughly evaluated and that any necessary adjustments or recommendations for future projects are documented.

- Data Analysis and Interpretation (Oct 2025 - Dec 2025)
- Final Reports and Recommendations (Jan 2026 - Mar 2026)

Key Milestones

- Completion of Feasibility Studies - Dec 2023
- First EOR Technique Deployed - Jun 2024
- First Significant Increase in Production - Dec 2024
- Final Project Report - Mar 2026

Benefits of the Project

If successful, the Enhanced Oil Recovery Project will provide numerous benefits:

- **Increased Oil Production:** Implementing advanced EOR techniques will significantly increase oil production from existing reserves.
- **Enhanced Energy Security:** Increasing the share of domestically produced oil will enhance Brazil's energy security and reduce dependence on imported oil.
- **Economic Growth:** The project will create jobs, stimulate local economies, and generate significant revenue through increased oil production.
- **Environmental Sustainability:** The project will promote environmental sustainability by reducing the environmental impact of oil extraction and production.
- **Social Development:** Providing stable and increased oil production will improve quality of life, reduce energy costs, and promote social development.

Risks and Mitigation Strategies

The Enhanced Oil Recovery Project involves several risks, including technical challenges, environmental impacts, and regulatory compliance. Given the high risk level, stringent safety protocols and comprehensive risk mitigation strategies will be implemented:

- **Technical Challenges:** Utilizing advanced EOR technology and experienced personnel to minimize technical risks. Regular training and drills will be conducted to ensure preparedness.
- **Environmental Impacts:** Conducting thorough environmental assessments and implementing best practices to minimize ecological damage. Regular monitoring and reporting will be implemented to ensure compliance with environmental regulations.
- **Regulatory Compliance:** Ensuring all activities comply with local and international regulations through regular audits and updates. Collaboration with regulatory bodies will be maintained to ensure ongoing compliance.

Safety Protocols

To ensure the safety of personnel and the environment, the following safety protocols will be implemented:

- **Emergency Response Plans:** Develop and implement comprehensive emergency response plans for potential incidents.
- **Safety Training:** Provide regular safety training to all personnel involved in the project.
- **Personal Protective Equipment (PPE):** Ensure all personnel are equipped with appropriate PPE and enforce its use.
- **Regular Inspections:** Conduct regular inspections of equipment and facilities to identify and mitigate potential hazards.
- **Incident Reporting and Analysis:** Establish a system for reporting and analyzing incidents to prevent recurrence.

Financial Benefits and Sharing

The Enhanced Oil Recovery Project is expected to yield a 22% Return on Investment (ROI). The financial benefits will be phased over time as follows:

- **Year 1 (2024):** Initial investment and setup costs.
- **Year 2 (2025):** Initial increase in production; start of revenue generation.
- **Year 3 (2026):** Significant increase in production; peak revenue generation.
- **Year 4 (2027):** Continued production and revenue generation; maintenance of increased production levels.

The financial benefits will be shared between BP and Petrobras based on their respective investments and contributions to the project. A detailed financial plan and agreement will be established to outline the sharing of costs and benefits.

Conclusion

The Enhanced Oil Recovery Project represents a significant opportunity for Brazil to enhance its oil production capacity, contribute to economic growth, and position itself as a leader in advanced oil recovery techniques. By carefully planning, executing, and monitoring the project, we can ensure its success and maximize its benefits.